

1 Two methods of estimating the water potential of plant tissue are described.

Method 1: Pieces of tissue are left in sucrose solutions of different concentrations. The percentage change in mass of the tissue is determined.

Method 2: Pieces of tissue are left in sucrose solutions of different concentrations. The change in the density of the solutions in which the pieces of tissue have been left is estimated.

A student compared the use of method 1 with method 2 to estimate the water potential of tissue in the fruit wall of red pepper, *Capsicum annum*.

Fig. 1.1 shows a red pepper fruit and a labelled longitudinal section of the same fruit.

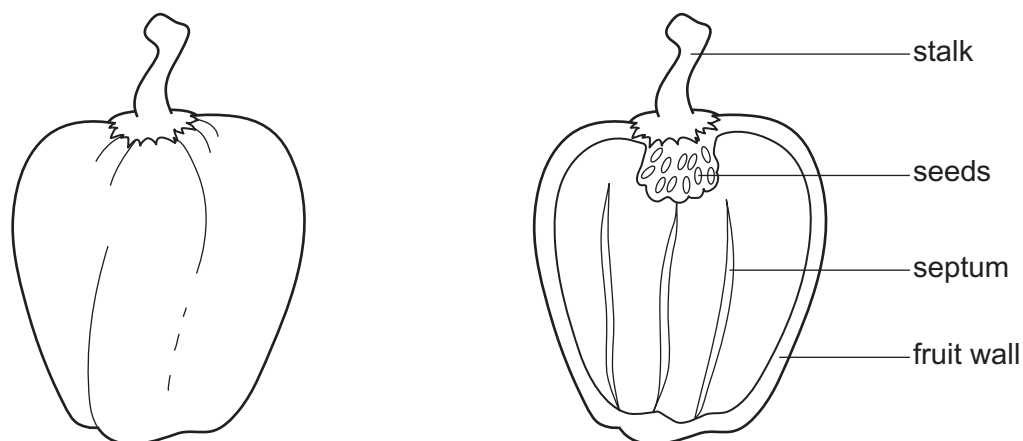


Fig. 1.1

The student first used method 1.

- Pieces of the red pepper fruit wall tissue of known mass were prepared.
- These pieces of tissue were placed in solutions of different sucrose concentration.
- The change in mass of the tissue after a period of time was calculated.

(a) (i) State the independent variable **and** the dependent variable in this investigation.

independent

dependent

[2]

- (ii) The student was given a 1.0 mol dm^{-3} stock solution of sucrose. The student decided to use the stock solution to make a range of sucrose solutions. The student made 50 cm^3 of each solution.

Suggest a suitable range of solutions the student could use **and** describe how these solutions could be made by **proportional** dilution of the 1.0 mol dm^{-3} stock solution of sucrose.

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- (b)** Using the different sucrose solutions from **(a)(ii)**, describe a method the student could use to collect the data needed to estimate the water potential of the tissue in the fruit wall of the red pepper.

Do **not** include details of how to make the different sucrose solutions.

Your method should be set out in a logical way and be detailed enough to let another person follow it.

[6]

- (c) (i) The student calculated the percentage change in mass of the tissue.

Describe how the student could have calculated the percentage change in mass of the tissue **and** explain why the percentage change should be used.

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- (ii) Sketch a graph of the expected results on Fig. 1.2.

Include axes labels and units.

Indicate how this graph could be used to estimate the sucrose concentration equivalent to the water potential of the tissue in the red pepper fruit wall.



Fig. 1.2

[3]

(d) The student then used method 2.

- Large test-tubes were set up in pairs.
 - Each pair of test-tubes contained the same volume of a concentration of sucrose solution.
 - The concentrations ranged from 0.1 mol dm^{-3} to 0.7 mol dm^{-3} .
1. Red pepper tissue was placed in one of the test-tubes containing 0.1 mol dm^{-3} sucrose solution.
 2. After the tissue had been soaking in the solution for a period of time, the tissue was removed.
 3. The solution from which the red pepper tissue had been removed was coloured by adding methylene blue solution.
 4. One drop of this coloured solution was then placed in the other large test-tube containing 0.1 mol dm^{-3} sucrose solution as shown in Fig. 1.3.

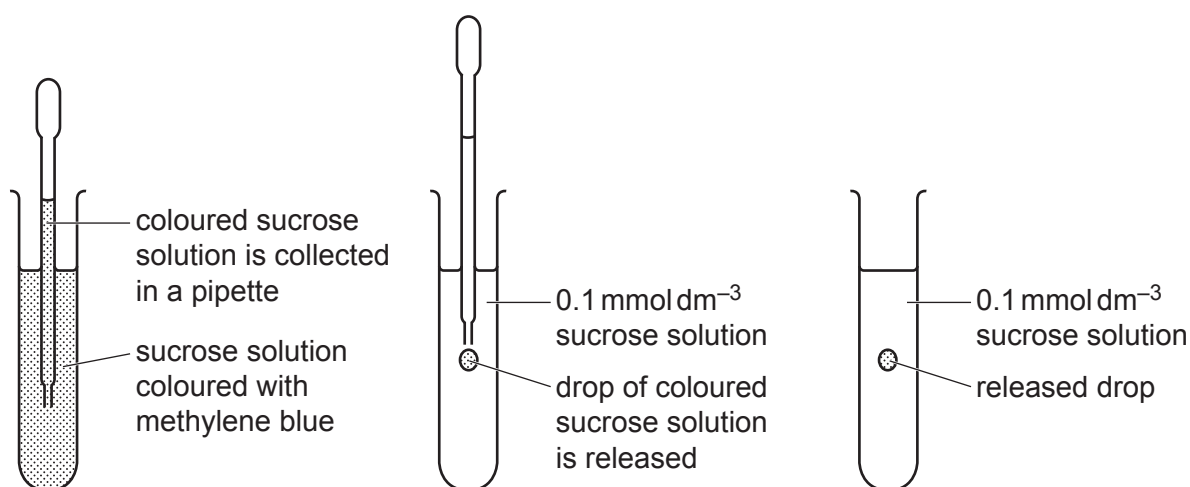


Fig 1.3

5. Step 1 to step 4 were repeated with the other concentrations of sucrose.
- If the plant tissue has gained water from the sucrose solution in which it has been soaking, the released drop will move down in the tube.
 - If the plant tissue has lost water to the solution in which it has been soaking, the drop of coloured solution will be less dense and the released drop will move up in the tube.

The results are shown in Fig. 1.4.

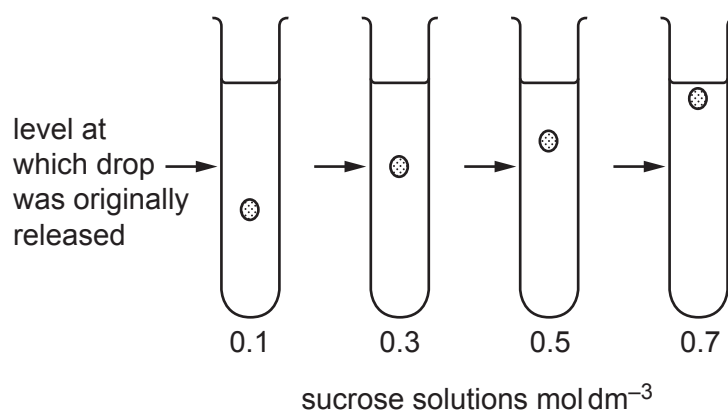


Fig. 1.4

Table 1.1 shows the water potentials of the sucrose solutions used by the student in method 1 and method 2.

Table 1.1

sucrose concentration / mol dm⁻³	water potential / kPa
0.10	-260
0.20	-540
0.30	-860
0.40	-1120
0.50	-1450
0.60	-1800
0.70	-2180

- (i) Using the results of the experiment in Fig. 1.4 and the data in Table 1.1, state the water potential of the red pepper tissue.

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- (ii) Suggest why the student thought that method 1 would give a better estimate of water potential than method 2.

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[Total: 20]